

metrics leveraged for analytics.

The real differentiation came in when the system got incorporated with a 3-axis gyroscope and 3-axis accelerometer during its development phase more than a year ago. As gyroscope uses Earth's gravity to help determine orientation, it gives orientation of movement of the pet dog. The accelerometer gives non-gravitational acceleration data whenever the dog walks or runs.

Software and analytics

The software, which is the soul of Wagr, has two parts. One runs on the smartphone and the other on the server-side analytics platform. Wagr utilises its smartphone application to co-ordinate with the wearable by syncing with it. The wearable sends data to the application, and the application talks to the server for feedback. You get graphs or reports on the smartphone screen just like an end-user report or a dashboard.

Analytics shows up patterns and dashboard primarily using inputs given by dog behaviourists and trainers, which are fed into the prebuilt evaluation measures in the database. The latter are updated every time a new behaviour or a subsequent change is seen by experts who know the dog's psychology.

The spec list

- Rechargeable lithium-ion-polymer batteries that last up to seven days
- Charging through Pogo pins
- 6-DOF motion sensor
- 3-axis accelerometer and 3-axis gyroscope
- GSM quad-band connectivity with pre-installed SIM card
- Bluetooth 2.1 SPP and 4.0 GATT (dual mode)
- Wi-Fi 802.11 b/g/n
- Positioning GPS/GLONASS/BeiDou
- Route tracking with real-time location update
- Rugged and waterproof
- Three multi-colour status indicating LEDs
- Compatible devices: Apple/iOS 7 and higher; iPhone 4S, 5, 5S, 6 and 6S, iPad, and Android 4.1 and higher

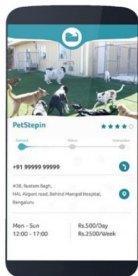


Fig. 2: Software screen for services

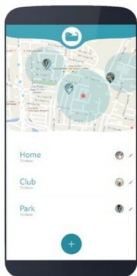


Fig. 3: Software screen for location tracking of safe zones



Fig. 4: Software screen for home activity monitoring

Mixing the above-said information with the analytics part in the background crunches data and analyses it against such parameters in real time. Apache Prediction IO is the tool of choice for the team; it is totally open source.

While dealing with analytical engines, challenges were encountered such as pushing and processing data to the analytics engine and trying to stream in data real time. It had to be based on data from animal movements and orientation, which is a time-consuming process. Once the prebuilt evaluation measure was set and data from wearables started flowing through priority based connectivity based on distance during the earlier days, the team tasted success.

The data now had to be represented in a meaningful yet interactive form, which the mobile application development team took care off. The application features a built-in map, a social networking platform, assistance services and a shopping cart. You can download this application and start syncing the wearable instantly, until the profile of the pet dog pops up on the smartphone screen saying "Hi."

Louder than words

Wagr's ecosystem has a photo-driven social platform that could be used



The team behind Wagr (L to R): Siddharth Darbha, Advait Mohan and Ajith Kochery

to access dog-related services such as vets, food, supplies, accessories, boarding, grooming, dog walkers and dog sitters. The platform brings active communities of dog-parents closer.

Leveraging this sharing capability through social community groups built into the application, long-term goals that its founders had in mind of co-existing with these lovable creatures could be fulfilled. These include adoption/rehabilitation, training tips and even rescues, all in real time.

Wagr is available across India through registration on its website (www.wagr.in). Thanks to the IoT and the value chain it creates, young and innovative technology community is finding new ways to add value to everyone who shares our planet. **EFY**